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What distinguishes binary from multi-class intrusion detection systems: Observations from experiments

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ABSTRACT

Modern world has become prune to technology and security is turning invasive by the day. Thus, capturing personal information or access to remote devices can prove to be horrendous intrusions. This paper focuses on various classification algorithms such as K-nearest neighbor Classifier, Multi Layer Perceptron Classifier, Long Short-Term Memory Classifier and Support Vector Machine Classifiers on the revised KDD cup 99 dataset. Attacks namely DoS (Denial of Service attacks), R2L (Root to Local attacks), U2R (User to Root attack) and Probe (Probing attacks) were monitored. Getting the model ready, we aim to identify the attack types based on the data coming through. The study also showcases Uni-variate, Bi-variate as well as Multivariate analysis on the same. The models were optimized and accuracy was found through measures like F1-score, precision, and recall. Promising results were found.

1. Introduction

Modern world is highly dependent on smart devices and the Internet. Communications can change people's attitudes and beliefs. The entire world strives on modern infrastructure and up-to-date amenities. As a result, efforts are being made all over the world to create smart cities. Chatterjee, Kar, & Gupta (2018) aids in understanding how the system security and confidentiality clauses fit best in residents' adoption of IT-enabled services in India's proposed smart cities. We have all heard that "Security is a myth these days". People all over the world present different perceptions of the word "security". The term ǣcommunicationǵ has numerous ramifications. The successful procedure of interaction between the people and their environment can be coined as communication. Through such interactions, two or more individuals are able to share their thoughts, theories as well as ideologies. Such communications are caused by the exchange of information through words, gestures, cryptograms, and expressions. With the advent of Web 2.0 Kar, Ilavarasan, Gupta, Janssen, & Kothari (2019), world became more and more technically compatible. Thus came the need to develop and nurture effective and operative communication mechanisms. Shahim (2021) showcases how Information Technology (IT) jargon coupled with the cash flow, make it difficult for citizens to fully understand security clauses and threats.

Ever, Sekeroglu, & Dimililer (2019) states that in the last few decades, artificial intelligence has progressed tremendously. AI has been able to outperform the everyday human jobs. Some of the problems of the digital era is related to networking and security tasks. The amount of data transferred every second has increased tremendously Within the last two decades. Machine learning techniques for detection, prevention and as well as classification have been developed for solving real-life problems. In organizations, communication is an essential process of giving and receiving information and to establish social relationships. As more people use the internet, the traffic increases Lazarevic, Ertoz, Kumar, Ozgur, & Srivastava (2003). Each web server can produce heavy traffic regardless of its number of clients. The traffic data can be analysed to classify it in normal and abnormal cases. Pertaining to the excess use of digital information, network traffic has increased rapidly in the last decade. The concept of transparent and user-friendly communication protocols was developed to make them more secure and easier to use. Due to the flexibility of the protocols, they can be easily manipulated by hackers Lee & Stolfo (1998). In order to safeguard systems from these attacks, intrusion detection is important. Artificial intelligence helps detect and even resolve issues related to web attacks.

Detection of various intrusion detection can be done using machine learning techniques such as Convolution Neural networks as described in Ding & Zhai (2018) or Hybrid Data Mining Approach in Bhattacharjee, Fujail, & Begum (2017) as well as using Vectorised Fitness Function in Genetic Algorithm in Parsaei, Rostami, & Javidan (2016). A fitness function is used to evaluate how optimal the proposed solution is to the desired problem statement. Genetic Algorithms contain numerous strings of numbers which when evaluated are assigned a fitness score. This score can also be termed as best/worst score. It helps define how fit a particular solution is. This paper aims to showcase a short comparison between various machine learning

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algorithms and how successfully they were able to detect intrusive attacks.

This study has been done to bridge the gap between the use of AI/ML algorithms and Intrusion detection systems (IDS) by collecting results based on various algorithms. The best parameters were set for respective algorithms to get the optimal solutions. This research was done with the following objectives in-mind:

1. To provide a detailed explanation on various types of attacks and importance of Intrusion detection Systems.
2. To define various Hypothesis and respective findings on Uni-variate and Bi-variate parameters to carry out Binary and Multi-class Classifications.
3. To explore different classification algorithms like Linear and Quadratic Support Vector Machine Classifier with Linear Kernel, K-nearest-neighbor Classifier and Long Short-Term Memory Classifier.
4. To create a framework as well as measure accuracies for classifying various types of attacks (Dos, Probe, R2L, U2R, normal) using K-nearest-neighbor Classifier and Multi Layer Perceptron Classifier.

This study has been organized to collate the following sections starting with the introduction in Section 1, followed by literature review in Section 2 which throws a light on Information systems, Cyber security threats as well as the Artificial Intelligent algorithms to aid them. Section 3 caters to Research methodology including an introduction to Uni-variate and Bi-variate analysis; followed by Findings in Section 4. A discussion based on practical and theoretical implications of the study has been mentioned in Section 5. The conclusions and future scope of the study can be found in Section 6.

2. Theoretical background

The world has emerged into a tech-driven domain. The drastic evolution from wired phones to the satellite TVs, all the way through SpaceX has changed information systems on an enormous scale. Information collection via various sources and domains proved to be useful for analytics and drawing out solutions for complex business problems. With this increasing demand in data, data security became more and more necessary. Today, preserving data and keeping it secure is one of the most important requirements of any organization. Micro to Macro, all organizations require the basic understanding and analysis of information systems Alter (1996). The term information can refer to anything and everything in nature. It includes objects, datasets, images, personal information, addresses and even likes and dislikes of a person Checkland & Holwell (1998). It can be based out of a book or even extended to or derived from multiple sources or views. It can be imaginary or theoretical. Taking an example of a library, the books and literature from it can be classified as information Reddy & Reddy (2014). The archives that store information about the books can be considered as information systems Kunz & Rittel (1970). There are several issues that can be associated with information systems such as goodness of the system, properties, retrieval systems, etc which have been discussed in Brancheau & Wetherbe (1987).

Cyber security challenges in information systems depend on various factors such as the integrity and authenticity of the information Elmaghraby & Losavio (2014), æprivacyg and rights to the information as well as services that are based on that information. With the increase in cyber crimes day-by-day the need of intrusion detection has become an utmost priority Tonge, Kasture, & Chaudhari (2013). Cyber Security extends not just to the IT domain but to every individual on the internet. Every domain based on data can become a prey to these intrusion attacks. Be it health care, communication, architectures or even food industry; attacks prevalent today can result into dangerous consequences Khan, Ibrahim, & Hussain (2021). Artificial intelligence has come a long way and is very closely related to Cyber security detection systems. Reshmi (2021) AI and ML have come to a point where intrusion detection can be done within a couple of seconds. Various algorithms and intrusion detection systems can be set up on-premise or even on cloud to aid these situations; thus protection data to its best potential Deepu & Ravi (2021).

At the start of the AI world, people started using the pattern matching algorithms in order to detect intrusion. Pattern matching algorithm Zhang (2009), Wu & Shen (2012) were being used in the context of feature matching. Many of the algorithms were carefully examined for the same purpose. Many algorithms such as KnuthMorrisPratt algorithm, BoyerMoore string-search algorithm, BMH algorithm, Aho-Corasick as well as AC-BM algorithm were implemented in Yin (2012). More accurate the results from the algorithms; more accurate will be the overall model implementation. In Dagar, Prakash, & Bhatia (2016), Naive approach, Knuth-MorrisPratt algorithm and RabinKarp Algorithm are associated to create a summary model for intrusion detection. Networks carry data in single packages. These packages contain information that helps analyze networks, predict network traffic, and even identify network performance issues. These files are saved with.pcap file extension. Pcap files can play an important role by detecting any network issues which might need immediate attention. Datasets even comprised of Pcap files as they helped increase the accuracy of the algorithms by 16

Network traffic is sky rocketing due to the use of smart devices. Some features prove to be redundant which result in a lengthy detection process Alrawashdeh & Purdy (2016). Performance evaluation was done with the features Correlation-based Feature Selection techniques, IG (Information Gain) and measuring the gain Chae, Jo, Choi, & Park (2013). Better feature selection methods by means of feature averaging of total and each class was proposed in Revathi & Malathi (2013). Pre-processing data is an essential step in knowledge discovery. Accurate and reliable results of data analysis are guaranteed by correctly prepared data. Paulauskas & Auskalnis (2017) Decision Trees, Random Forest and Rule Based classifiers are used in the paper to analyse attack detection accuracy.

Poonam et al. Kumar, Chauhan, & Panwar (2013) made use of the outlier detection efficiency. In order to detect outliers while concurrently performing clustering operation; a state-of the art model for intrusion detection was found. Denatious et al. Denatious & John (2012) offers a study on data mining techniques. They help develop an effective identification models and well as attacks. Thus, the user can develop secure information systems. Efficient dataset selection as well as optimal features determine how easily attacks can be detected based on the right optimizers and learning rate Imran, Abdullah, Hussain, Palaniappan, & Ahmad (2012).

Optimizers play an important role with respect to performance. That being said, AdaBoost-based models were discussed in the paper Hu, Hu, & Maybank (2008). Logistic Model provides high detection rates and performance. It makes use of the cross validation to avoid overfitting of the data. One of the most common text categorization methods is k-Nearest Neighbor (kNN) classifier. It can be easily used to identify common attacks. It has been experimentally shown that KNN helps decrease the false positive rate, thus increasing the overall model accuracy. As KNN is more computationally advanced, it helps classify features by individual introspect. Liao & Vemuri (2002) showcases that kNN classier can effectively detect intrusive attacks.

The drawbacks of KDD dataset leading to new version of NSL KDD dataset can be found in Ingre & Yadav (2015) and Tavallaee, Bagheri, Lu, & Ghorbani (2009). Su, Sun, Zhu, Wang, & Li (2020) BAT model is basically a traffic anomaly detection model. It syndicates attention mechanism and long-term memory. The significant features for system traffic sorting can be obtained with the help of attention mechanism. The layered structure helps with the accumulation of local traffic data. The proposed model in Lakhina, Joseph, & Verma (2010) gives improved depiction of the data to reduce features. This increased about 80.4% data reduction, as well as 40% less required training time. The testing was even 70% faster. BAT performed better than the existing CNN and RNN

models. The need to develop a better dataset was due to the following mentioned reasons:

- KDD included redundant records in the training set. This resulted in classifiers being biased.
- Sufficient number of records were made available which helped in increasing the accuracy of the model. This also helped make the study reasonably affordable without specifically targeting a single section of dataset.
- Frequent records can cause model to attain higher accuracy. This can be termed as a biased behaviour. To aid this error, duplicate records were removed from the old dataset to help create the new NSL-KDD dataset.
- The carefully selected records in the new dataset are inclusive to each and every difficulty level group. The newly devised records are inversely proportional to the percentage of original KDD data set records.
- With multiple varying range of classification models being performed on the dataset, new and better accuracy models can be curated.
- Subsequently, the average results of different research works will be open for further research.

3. Research methodology

3.1. Dataset description

The KDD Cup 1999 Dataset was created for the task of building a network intrusion detector. Meena & Choudhary (2017) Numerous records based on the traffic prevailing online were composed and bundled into a data set called the KDD'99. The main purpose was to create an analytical predictive model proficient in distinguishing good connections from the bad ones. The "bad" connections can also be termed as intrusions or attacks.Gurung, Ghose, & Subedi (2019) The "good" connections are termed as normal connections. This database constituted of a typical dataset. It included diverse types of intrusions simulated in a military network environment. NSL-KDD is a dataset recommended to address the characteristic problems related to previous data set. The NSL-KDD dataset was fetched into existence by the University of New Brunswick. It is one of the most famous and dependable datasets capturing modern day internet traffic. This dataset used in our study comprises of four sub datasets as shown below:

1. KDDTrain + .ARFF : It contains binary labels in ARFF format along with the training set.
2. KDDTrain + .TXT : It includes attack-type labels along with their respective difficulty levels.
3. KDDTrain + _20Percent.ARFF : A 20% subset of the KDDTrain + .arff file.
4. KDDTrain + /_20Percent.TXT : A 20% subset of the KDDTrain + .txt file.
5. KDDTest + .ARFF : It includes additional binary labels in ARFF format along with the Test set.
6. KDDTest + .TXT : It includes attack-type labels along with their respective difficulty levels for testing.
7. KDDTest-21.ARFF : It does not include records with difficulty level of 21 out of 21.
8. KDDTest-21.TXT : It excludes the records with a difficulty level of 21 out of 21.

KDDTest-21 and KDDTrain + 20Percent are subcategories of the KDDTrain + and KDDTest + datasets. Pertaining to the case; KDDTrain + will be referred to as train and KDDTest + will be referred to as test. The higher the Score of the column, the higher will be the traffic.

3.2. Various classes of attacks

Boyce et al. (2011) showcased the importance of security compliance, risk reduction and human error mitigation by outlining cost effective human-centered configuration of cybersecurity systems. NSL-KDD dataset comprises of 5 classes that are normal. It also entails 41 features based on traffic inputs, Score (how severe the traffic is) and 4 types of attacks. With three types of categories namely Nominal, Binomial and Numeric; the features were bifurcated. The protocol type, Services and Flags made up the Nominal block where as features like logging-in information, root shell, etc made the Binomial block. The data-set contained internet traffic encountered with the help of a simple intrusion detection network. The last two labels were assigned as 'normal' and 'attack' along with Score' indicating the importance of the traffic.

The four major categories of multi-class attacks found in this study are as follows Dhanabal & Shantharajah (2015):

- DOS refers to denial-of-service. This type of attack is responsible for completing stopping the traffic flow altogether. Thus no traffic will be able to get through to the network. For example, when there are big sales at Amazon/Flipkart or any e-commerce company; sometimes the status of the items don't get updated. This may result in people having sold out items in their carts during checkout. Attacks like mailbomb, neptune, etc are DOS attacks.
- Remote to user (R2L) refers to an attack that tries to acquire unauthorized access from a remote machine. Example - guessing password for websites and even CVVs for credit and debit cards, Satan, warezmaster, etc.
- User to Root (U2R) Attack tries capturing unauthorized access to the network and all the data contained within it. It tries to become the sole main user in order to exploit details to make profit. Example- various "buffer overflow" attacks, Perl, sqlattack, etc.
- Probing refers to surveillance and other probing. This attack tries to capture information and as much data possible out of the secure networks. The information can range from small details like a person's name and gender to credit details and even passwords, portsweep,saint, etc.

3.3. Feature selection

In order to get the most accurate results; feature selection was prioritized. Models trained on significant features become the most reliable model to classify test sets. Four types of classification was done which is showcased in the table below:

Chi Square was used to select features for binary classification. The steps followed were as described below:

Step 1- Define Hypothesis. Null Hypothesis (H0): Two variables are independent. Alternate Hypothesis (H1): Two variables are not independent.

Step 2- Curate the Contingency table. The samples that used tcp protocol type and led to intrusion and the samples that led to normal or no attacks.

Step3- Expected values were calculated. Correlation was found and Chi-square statistics were calculated to curate the best 20 features affecting the attacks.

3.4. Data standardization

Normalization of the dataset. The dataset is normalized in order to avoid over-fitting and missing values. Normalization helps standardize the dataset so that whenever any sort of misconduct occurs, it can be avoided easily. The base scale to standardize any continuous type of data is between the range of [0,1]. Normalization also helps with avoiding inconsistency while assigning weights to the parameters. Feature reduction applies a mapping of the multidimensional space into

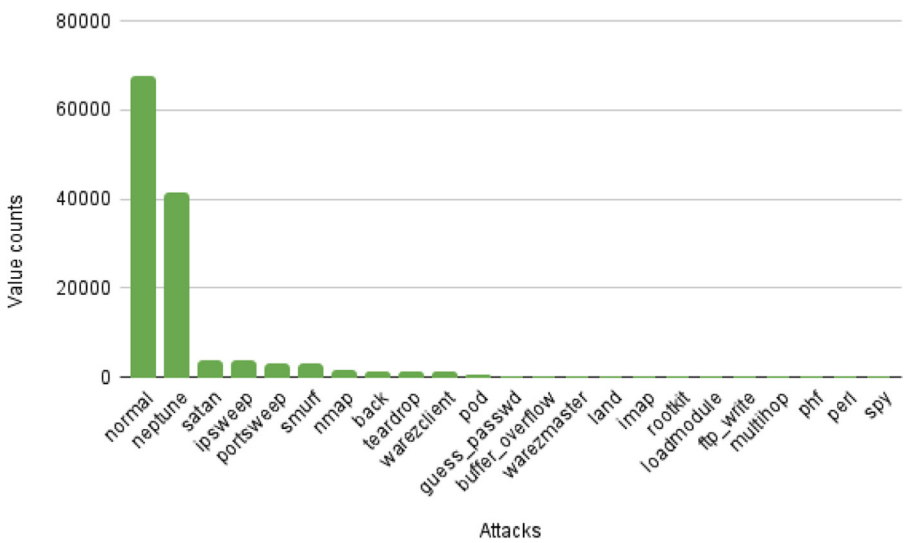

Fig. 1. Skewed Attacks.

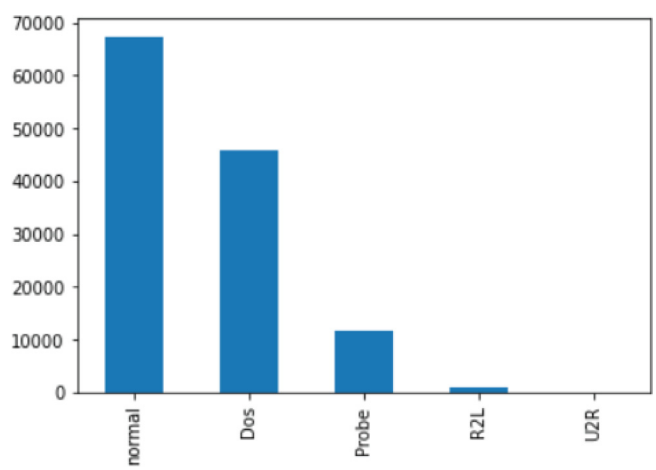

Fig. 2. Distribution of attack classes.

a space of lower dimensions. Correlation Feature Selection was used in Ingre, Yadav, & Soni (2017). The results stated that most of the outcome for binary class classification was more than that of the five attack classification.

3.5. Uni-variate data analysis

A categorical diversion of sub-classes for each attack has been showcased in Fig. 1.

Figure 1 depicts how all the attacks exist in the dataset are heavily uneven in their distribution.This shows a near-to-accurate representation of the world's internet traffic today. Most of the attacks are DoS(denial-of-service) and Probe; whereas U2R and R2L are hardly ever seen. Most of the records available in dataset are that of normal traffic.

Assigning labels to each of these attacks; 'Normal','DoS', 'Probe', 'Privilege' and 'Access', a cross-tab of Attack versus Protocol was obtained. Using the normal' attack as base; a network traffic analysis is done. This helps classify the attack classes easily which resulted in Fig. 2. This also helps us to find out which attack targets which specific protocol Kumar et al. (2013). Similarly, uniariate analysis of protocol type proved tcp protocol to be the most used protocol; followed by udp and lastly icmp. For the services tab, http was the most frequent followed by private, domain_u, smtp and ftp_data.

Table 1
Features Classification.

Classification	Feature Selection
Binary	numeric features with binary labels. encoded non numeric features like services, protocol, flag) with binary labels.
Multi-class	numeric features with multi-class labels. encoded non numeric features like services, protocol and flag) with multi-class labels.

Table 2
Bivariate Analysis.

Attack Class	Protocols : Number of samples
DoS	tcp: 42188 icmp: 2847 udp: 892
Probe	tcp: 5857 icmp: 4135 udp: 1664
R2L	tcp: 995
U2R	tcp :49 udp :3

3.6. Bi-variate analysis

Bi-variate analysis was done on Protocol and Attack Types which resulted in categorizing the number of samples in Dataset based on the protocol and attack type. This has been presented in detail in Table 2. Similar analysis was done on Service, Attack Types as well as Protocol and Services. Protocols used in the connection can be seen in Fig. 2 and Fig. 3. The Protocol Type and Service values define the connection. The Flag feature defines the standing of the connection, and whether a flag was raised or not. Each value in the Flag showcases status a connection had.

Charts were created to analyse the differences in each protocol type and to identify the type of traffic being generated. Coming to the services part, it can be noted that most of the normal traffic is http. Coupling the protocols, services, flags along with amount of data in the connection can help the administrator to analyse the data effectively. Modern network equipment includes various numeric data such as duration, src_bytes, dst_bytes along with protocol_type, service and flags.

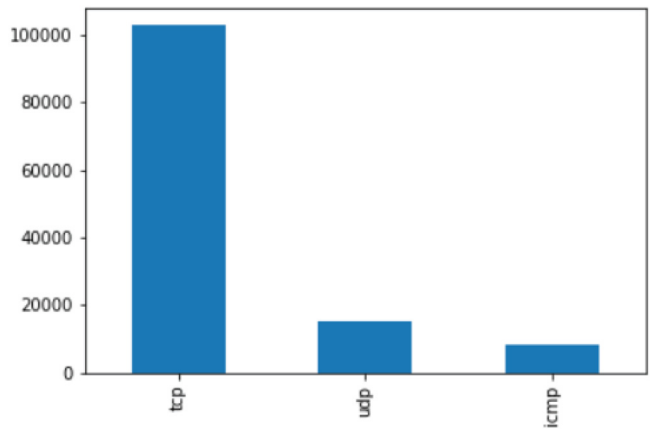

Fig. 3. Distribution of protocol classes.

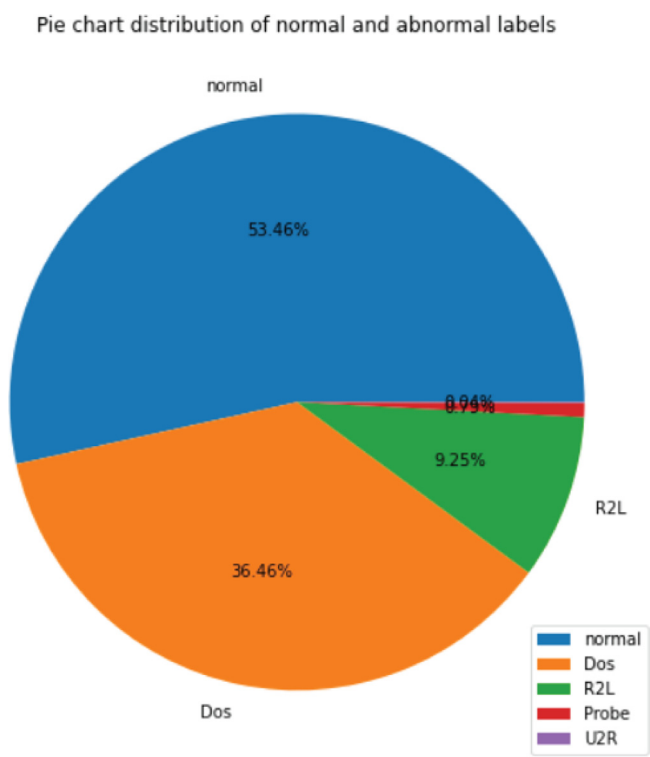

Fig. 4. Pie chart of Binary Distribution.

4. Findings

In order to gather analytical data from our results, the below mentioned analysis was done.

4.1. Binary classification

The binary classification comprises of two categories 'normal' and 'abnormal'. Label encoding was done assigning '0' to abnormal and '1' to normal. One-hot-encoding on attack labels helped to map and categorize attacks. In order for the computer to recognize data being passed to it; one hot encoding plays a vital role. It transforms data into binary code required for the computer's understanding. The pie chart distribution of normal and abnormal labels has been shown in Fig. 4.

To find the most correlation among features; Pearson correlation was used for binary class dataset. Pearson correlation is used to establish association between various continuous variables. In this case; (Intrusion-> yes(1) / no(0)). Thus by finding the attributes with higher than 0.5 correlation and by joining encoded, one-hot-encoded, and original attack label attribute; new dataset was curated for further analysis.

4.1.1. Linear and quadratic support vector machine classifier with linear kernel (Binary Classification)

Linear Support Vector Machine works really well on separating data into different classes. A straight line is used to draw a separation between the two classes; normal and abnormal. As this data was linearly separable, a linear kernel was used. LSVM-Classifier Binary Set-Accuracy was found to be 97.53%.

Quadratic programming (QP) is used to find a linear separation between datasets which might not be that linearly separable on their own as shown in Fig. 5. The x-axis showcases the records in NSD-KDD dataset whereas the y-axis showcases the binary labels assigned to the attacks; in this case 0 and 1. The purple lines depict the predictions drawn whereas the red lines showcase the real values of the attacks. QSVM helps to set in a quadratic optimized setting even in the presence of linear variables. QSVM-Classifier Binary Set-Accuracy was found to be 96.77%.

4.1.2. K-nearest-neighbor classifier (Binary Classification)

In order to find the most common features between the normal and abnormal classes; KNN was implemented.The model was created on 5 neighbors. KNN-Classifier Binary Set-Accuracy was found to 99.60%. The Fig. 6 depicts the graph obtained through this classifier. The x-axis showcases the records in NSD-KDD dataset whereas the y-axis showcases the binary labels assigned to the attacks; in this case 0 and 1. The purple lines depict the predictions drawn whereas the red lines showcase the real values of the attacks. The predicted values and real values almost overlap each other; justifying the 99.6% accuracy.

4.1.3. Long short-term memory classifier (Binary Classification)

A sequential model was created to predict the two classes using LSTM Classifier. With past history of models and their parameters as well as time taken to run each model; this classifier proves its excellence by adjusting weights.Activation function determines if the neuron should be activated or not; by adjusting weights and non-linearity into the output of the neuron. Sigmoid was used as the activation function as it is transforms the output into a value between 0.0 and 1.0. Adam was used to optimize the model. In order to find out and adjust the difference between the actual output and the predicted outputs, loss functions are used. In order for the model to resist the noise and produce good accuracy models; binary cross-entropy was defined as the loss function. As this is a linear classification; binary cross entropy was the optimal choice. Thus by defining loss function, optimizer, metrics and then compiling model the accuracy found was 97.48%. A graph plot of Accuracy Vs Epoch can be seen in Fig. 7. The accuracy clearly seems to be increasing with the increase in epochs.

4.2. Multi-class classification

A framework with multi-class labels (Dos, Probe, R2L, U2R, normal) was created. After label encoding and one-hot-encoding attack labels;a data frame comprising of just numeric features with multi labels was ready for training. In order to find the most correlation between features; Pearson correlation was used for multi-class dataset. Pearson correlation is used to establish association between various continuous variables. In this case; numeric features and multi-class (Types of Attacks). Thus by finding the attributes with higher than 0.5 correlation and by joining encoded, one-hot-encoded, and original attack label attribute; new dataset was curated for further analysis.

4.2.1. K-nearest-neighbor classifier (Multi-class Classification)

As KNN Tiwari, Palivela, & Kumar (2022) makes use of Euclidean distance, Manhattan distance, Majority vote, the model was easy to optimize and work with. Each model has it's specific requirements to produce the best solution accuracy. The number of neighbors is

Fig. 5. QSVM-Classifier Binary Set.

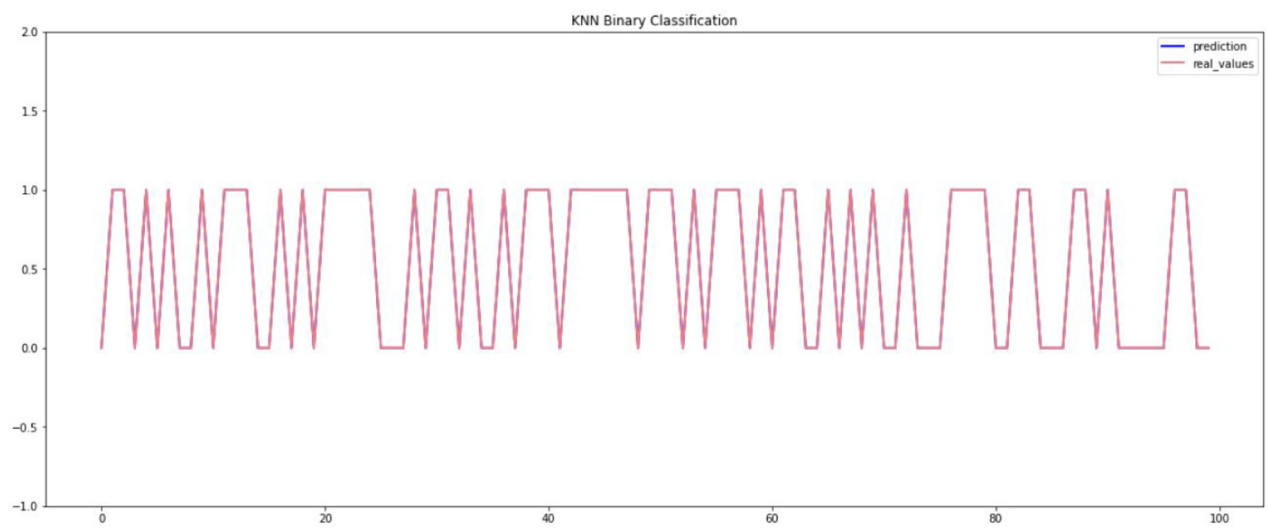

Fig. 6. KNN-Classifier Binary Set.

an accuracy controlling variable in KNN. Small number of neighbors make the model noisy whereas higher number becomes expensive to run and can also tamper with the accuracy. The proposed model was run for 5 neighbors with the training size of 0.25 which was chosen by Elbow method and lots of trial and errors. The model was very well able to predict different classes namely; DOS, Probe, R2L and U2L.The KNN-Classifier Multi-class Set-Accuracy was found to be 99.57%.

4.2.2. Multi layer perceptron classifier (Multi-class Classification)

Neural networks play an import part in the world of Deep Learning. Neuron was basically the atom of Deep Learning. It is like the core of our minds. It helps the other parts of the brain know if the information is worth processing or not. Perceptron can be thought of as an algorithm which helps detect things or solve things. For a binary classification, a linear hyper-plane can be created to find the exact boundary of the classes. Coming to multi-class classification, hidden layers become an essential part. For the proposed sequential model, Adam was used as the optimizer with soft-max activation function and categorical cross-entropy for loss function. The model was trained for 100 epochs. The overall accuracy was found to be 99.23%. Fig. 8 showcases the plot between accuracy and epoch of the same.

5. Discussion

5.1. Implications of this study

There are various implications based on this study which are both theoretical and practical. It was made sure that the study was done on the most apt and accurate dataset possible for intrusion detection. This study has been able to provide a detailed overview of various Machine Learning algorithms with respect to cyber-security domain. KDD Cup 1999 was the dataset used which aided building network intrusion detector. Various classes of Attacks namely denial-of-service, Remote to user (R2L), User to Root (U2R) as well as Probing were to be detected. Significant features were extracted for binary as well as multi-class classification of the above mentioned attacks. The results of this study showcase the importance of Intrusion detection for information systems using various machine learning algorithms at base. Findings were based on Linear and Quadratic Support Vector Machine Classifier with Linear Kernel, Long Short-Term Memory Classifier and K-nearest-neighbor

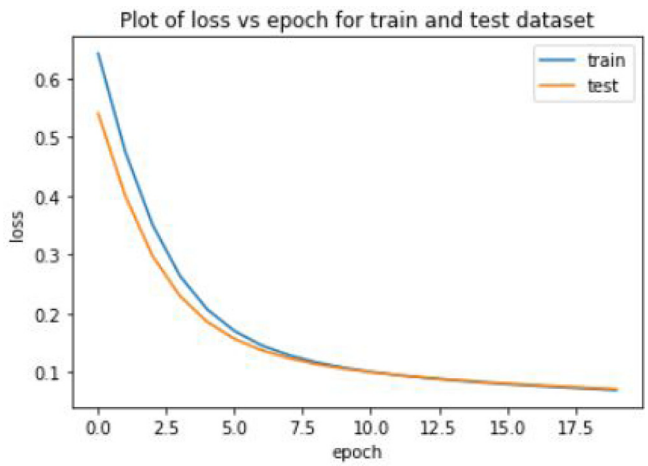

Fig. 7. Long Short-Term Memory Classifier.

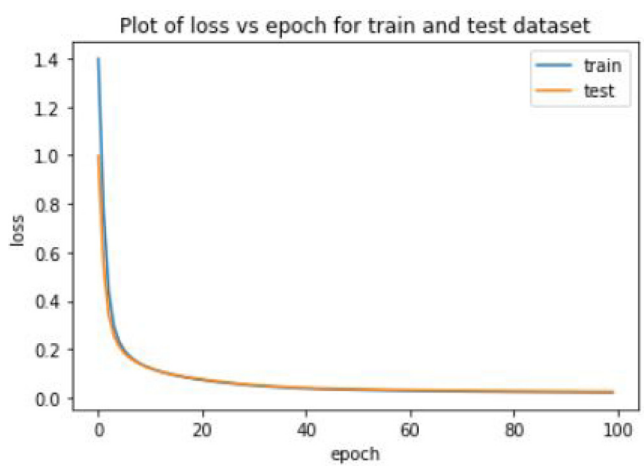

Fig. 8. Multi Layer Perceptron Classifier.

Classifier for Binary classification of attacks. Whereas Multi-class Classification was done on Multi Layer Perceptron Classifier and K-nearest-neighbor algorithms. In terms of practical implications, results can be used to create new IDS systems with optimal accuracies and can even better the existing systems. The findings of this study also draw a comparative analysis on various algorithms available for minimizing the risk of faulty detection.

5.2. Future research direction

Mustafa, Kar, & Janssen (2020) emphasizes on the importance of failures in information systems (IS) literature. The results imply that failures in information, functionality, systems, and services have an effect on consumers through various factors like operations, expenses, as well as satisfaction of digital services, these can be further studied. Innovative ensemble learning approaches can be used to create enhanced models for intrusion detection. The analysis in this study is limited to NSL-KDD dataset but can be extended to different use-cases in real world. Future research can explore that along with various deep learning algorithms.

6. Conclusion and future work

Various classification algorithms such as K-nearest neighbor Classifier, Multi Layer Perceptron Classifier, Long Short-Term Memory Classifier and Support Vector Machine Classifiers were used for the proposed study. The research was undertaken with a bifurcating approach showcasing Uni-variate, Bi-variate as well as Multivariate analysis. With Binary classification, the goal was to separate normal and abnormal classes. Whereas Multi-class classification shed a light on classifying various types of attacks such as denial-of-service, Remote to user, User to Root (U2R) and Probing. In order to calculate the accuracy of the model; various accuracy measures like F1-score, precision, and recall were undertaken. Promising results were found. Furthermore, in-depth analysis on feature selection and normalization can be undertaken.

CRediT authorship contribution statement

Aditya Palshikar: Conceptualization, Methodology, Software.

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